



RavenStack

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1. Executive Summary

This project developed an **end-to-end Customer Churn and Retention Intelligence framework** for RavenStack, a fictional subscription-based SaaS company. The goal was to move beyond basic churn reporting and build a unified analytical view that explains why customers leave, what risk signals appear before churn, and which customer segments should be prioritized for retention action.

Using a multi-table synthetic SaaS dataset, the project integrated account, subscription, product usage, support, and churn-event data into a structured analytical workflow. The project covered data preparation, data quality validation, exploratory analysis, feature engineering, and baseline predictive modeling at both the account and subscription levels.

The final outcome showed that the dataset supports **strong descriptive and diagnostic churn analysis**, but that predictive performance remained weak in the current feature space. As a result, the project's main value lies in providing RavenStack with a practical **retention intelligence framework** rather than a production-ready churn prediction model.

2. Business Context

RavenStack operates as a subscription-based B2B SaaS business, where recurring revenue depends on long-term customer retention. In this type of model, churn directly affects MRR, ARR, expansion potential, and customer lifetime value.

Customer behavior in RavenStack is distributed across multiple lifecycle stages, including account creation, subscription management, product usage, support interaction, and churn events. Because these signals are stored across separate relational tables, understanding churn requires more than isolated reporting—it requires an integrated analytical view of the customer journey.

This project addresses that business need by connecting product engagement, subscription structure, support experience, and churn outcomes into a unified retention analysis framework.



3. Problem Statement

RavenStack lacks a unified analytical framework that explains churn across the full subscription lifecycle. Although the business captures customer data across accounts, subscription, usage, support, and churn-event records, these signals are not yet connected into a single decision-oriented view.

As a result, the business can observe churn after it happens, but cannot reliably explain why it happened, which customers were most vulnerable, or what warning signs appeared before attrition occurred. This keeps retention decisions reactive rather than proactive.

The core problem is therefore not only customer loss itself, but the absence of a structured analytical way to identify the commercial, behavioral, and operational patterns associated with churn risk.

4. Project Objectives

The main objective of this project is to build a **Customer Churn and Retention Intelligence framework** that explains churn through an integrated view of:

- account profile
- subscription lifecycle
- product usage behavior
- support experience
- churn outcomes

The project specifically aims to:

- identify the strongest churn-related patterns across commercial, behavioral, and operational dimensions.
 - detect early warning signals linked to customer attrition.
 - support customer segmentation for retention prioritization.
 - measure revenue exposure associated with churn.
 - translate analytical findings into practical business recommendations.
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5. Stakeholders / Business Users

This project is relevant to multiple business functions across RavenStack:

- **Executive Leadership** — to understand churn trends, revenue risk, and retention priorities.
 - **Product Teams** — to evaluate whether feature adoption and product friction are linked to churn.
 - **Customer Success Teams** — to identify at-risk customers and prioritize proactive intervention.
 - **Support Teams** — to assess whether support burden, escalation, or satisfaction patterns relate to attrition.
 - **Revenue / Growth Teams** — to understand MRR/ARR exposure and retention opportunities.
 - **Data & Analytics Teams** — to extend the work into dashboards, monitoring, or future predictive modeling.
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6. Data Source

The project uses the **RavenStack synthetic SaaS dataset**, a multi-table CSV dataset designed to simulate the customer lifecycle of a subscription-based B2B SaaS company.

The dataset includes five related files:

- ravenstack_accounts.csv
- ravenstack_subscriptions.csv
- ravenstack_feature_usage.csv
- ravenstack_support_tickets.csv
- ravenstack_churn_events.csv

Because the dataset is synthetic, it contains no real customer information while still preserving the relational and behavioral structure needed for analytics and modeling.



7. Data Understanding

The RavenStack dataset is a **relational multi-table SaaS dataset** built to represent the full customer lifecycle. Instead of storing customer data in one flat table, the dataset separates:

- account information
- subscription history
- feature usage behavior
- support activity
- churn events

This structure is analytically valuable because churn is not created by one business layer alone. It is shaped by the interaction between customer profile, revenue structure, product engagement, support experience, and churn outcomes.

The dataset is linked primarily through two keys:

- account_id
- subscription_id

This relational design made it possible to build both **account-level** and **subscription-level** analytical views later in the project.

8. Methodology

The project followed an end-to-end analytics workflow designed to convert relational SaaS data into a structured churn and retention intelligence framework.

The workflow included:

1. **Data loading and structural understanding**
2. **SQL-based preparation and relational integration**
3. **Data quality assessment and cleaning**
4. **Feature engineering**
5. **Exploratory data analysis (EDA)**
6. **Baseline predictive modeling and evaluation**

This methodology was built around one principle: churn should be analyzed as a **business process**, not only as a final binary label. For that reason, the project combined commercial, behavioral, support, and churn-event signals into one structured analytical process.



9. SQL-Based Data Preparation and Analytical Layer

SQL played a foundational role in the project because the RavenStack dataset is relational by design. Its key role was to transform separate raw CSV tables into a structured analytical layer before Python-based analysis began.

The SQL stage supported:

- raw data ingestion
- schema and type standardization
- relational joins across account, subscription, usage, support, and churn tables
- aggregation of event-level records into customer-level and subscription-level analytical metrics

This SQL preparation layer reduced downstream complexity and established a reliable analytical base for data quality review, exploratory analysis, and feature engineering.

10. Data Quality Assessment and Cleaning

The data quality stage confirmed that the RavenStack dataset was structurally sound, relationally consistent, and analytically ready for downstream analysis.

The validation covered:

- completeness
- uniqueness
- referential integrity
- temporal consistency
- categorical and numeric validity

The main findings were:

- no critical orphan records across the relational model
- no blocking duplicate-key issues
- corrected temporal parsing for major date and datetime fields.
- limited missing values, mostly business-expected.
- valid numeric ranges across revenue, support, and usage measures



Only a few edge cases required interpretation rather than deletion, such as:

- satisfaction_score = 0
- zero-revenue trial subscriptions
- rare zero-usage records

Overall, the dataset was confirmed to be clean and reliable enough for exploratory analysis and feature engineering.

11. Exploratory Data Analysis (EDA)

The EDA stage examined churn from multiple business perspectives rather than treating it as a standalone outcome. The analysis covered:

- **account and subscription structure.**
- **revenue and billing**
- **product usage**
- **support experience**
- **churn events**
- **cross-domain behavioral patterns**

The main analytical findings were:

- RavenStack has a diversified subscription portfolio with clear variation across plan tiers and revenue contribution.
- zero-revenue subscriptions were fully explained by trial status.
- Product usage showed meaningful variation in engagement depth and feature adoption.
- Product friction was concentrated in selected features rather than spread uniformly.
- Support burden was analytically relevant, but not a strong standalone churn driver.
- Churn reasons were multi-causal, with product, support, budget, pricing, and competitor factors all present
- No single dominant churn driver emerged across the business.

The most important conclusion of the EDA stage was that churn in RavenStack behaves as a **multi-factor business outcome** rather than a one-variable problem. This made feature engineering and structured churn modeling the natural next phase of the project.



12. Feature Engineering for Churn Analysis

After completing the exploratory analysis, the project moved into feature engineering to transform the cleaned RavenStack dataset into a more useful analytical structure for churn assessment.

Two feature-engineering paths were developed:

- an **account-level pipeline**, where customer behavior was aggregated at the account level.
- a **subscription-level pipeline**, where each subscription was treated as a separate analytical unit.

Across both paths, engineered variables were created to better represent:

- product engagement
- usage intensity
- feature adoption
- product friction and error burden
- support exposure
- revenue and commercial value
- churn-history context.

Additional steps were also applied to improve feature quality, including:

- missing-value review
- Skewness treatment
- leakage-aware feature removal
- redundancy reduction
- and class-imbalance handling

Overall, the feature-engineering stage successfully transformed the relational RavenStack dataset into structured analytical feature sets suitable for baseline churn modeling.



13. Modeling Assessment

Baseline predictive modeling was tested using both:

- **Logistic Regression**
- **Random Forest**

These models were evaluated at both the **account level** and the **subscription level**, with additional attention to:

- target leakage
- class imbalance
- threshold tuning
- and feature refinement

Although the modeling workflow was technically successful, predictive performance remained weak across the tested baselines. No model produced strong and reliable churn classification performance, even after:

- leakage removal
- feature improvement
- imbalance handling
- and analytical-unit reframing

This indicates that the current dataset supports **descriptive and diagnostic churn analysis** more strongly than **high-confidence predictive churn modeling** in its present form.



14. Business Recommendations

Based on the project findings, RavenStack can already benefit from the analysis in several practical ways, even without a strong predictive model.

Recommended actions include:

- monitor churns through a **multi-dimensional retention framework** rather than through a single churn flag.
 - prioritize operational review of:
 - product friction
 - support burden
 - escalation signals
 - trial and low-value subscription patterns
 - use the **subscription-level analytical view** for more granular portfolio review and risk segmentation.
 - separate **trial / zero-revenue subscriptions** from paid subscriptions when making retention decisions.
 - treat the current analytical framework as a **retention intelligence system**, not yet as a production-level churn prediction engine.
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15. Project Limitations and Future Work

The main limitation of the project is that the available churn signal in the current dataset remained weak for predictive modeling, both at the account and subscription levels.

While the project included strong feature engineering and multiple modeling attempts, the dataset still appears to lack the kind of deeper temporal and behavioral signals often required for strong churn prediction in SaaS environments.

Recommended future improvements include:

- adding **recency and tenure features**
- building **trend-based usage and support features**
- incorporating **behavior over time**, not only aggregated totals
- including **customer-success intervention history**
- testing more advanced models only after richer feature enrichment

In other words, future performance gains are more likely to come from **better data representation** than from simply trying more baseline models.



16. Final Project Conclusion

This project developed an end-to-end **Customer Churn and Retention Intelligence framework** for RavenStack by integrating account data, subscription structure, usage behavior, support activity, and churn-event context into a unified analytical workflow.

The project successfully covered:

- data preparation and analytical structuring
- data quality assessment and cleaning
- exploratory analysis
- feature engineering
- baseline predictive modeling
- modeling evaluation and recommendation

The final outcome is valuable not because it produced a strong churn classifier, but because it clearly demonstrated:

- how churn can be analyzed across multiple business dimensions
- which behavioral and operational patterns are most relevant
- and what limitations currently prevent stronger predictive performance

As a result, the project provides RavenStack with a strong **analytical foundation for retention intelligence**, while also clarifying what additional data and feature depth would be needed for future predictive improvement.



17. Power BI Dashboard and Reporting Layer

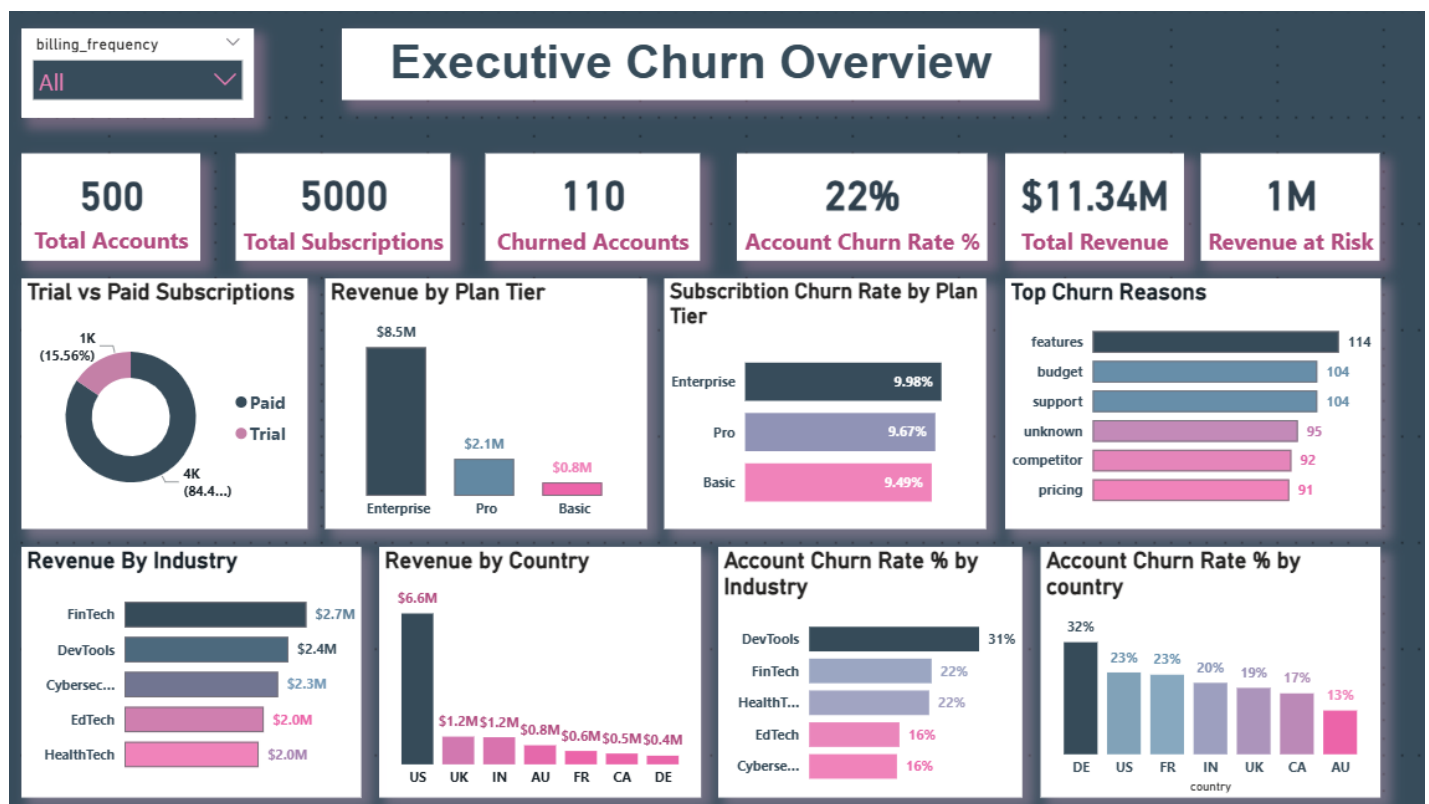
To complement the analytical and modeling stages of the project, a Power BI reporting layer was developed to translate the main findings into an interactive business-facing format. The purpose of this layer is not to replace the underlying analysis, but to make the project's key insights easier to explore, communicate, and operationalize for non-technical stakeholders.

The dashboard component was designed to support the practical use of the churn and retention framework by presenting the most relevant business indicators in a clear and accessible way. While the Python-based workflow focused on data preparation, feature engineering, exploratory analysis, and modeling assessment, the Power BI layer serves as the visual decision-support interface of the project.

The reporting design was structured around three business-oriented dashboard pages:

- **Executive Churn Overview**

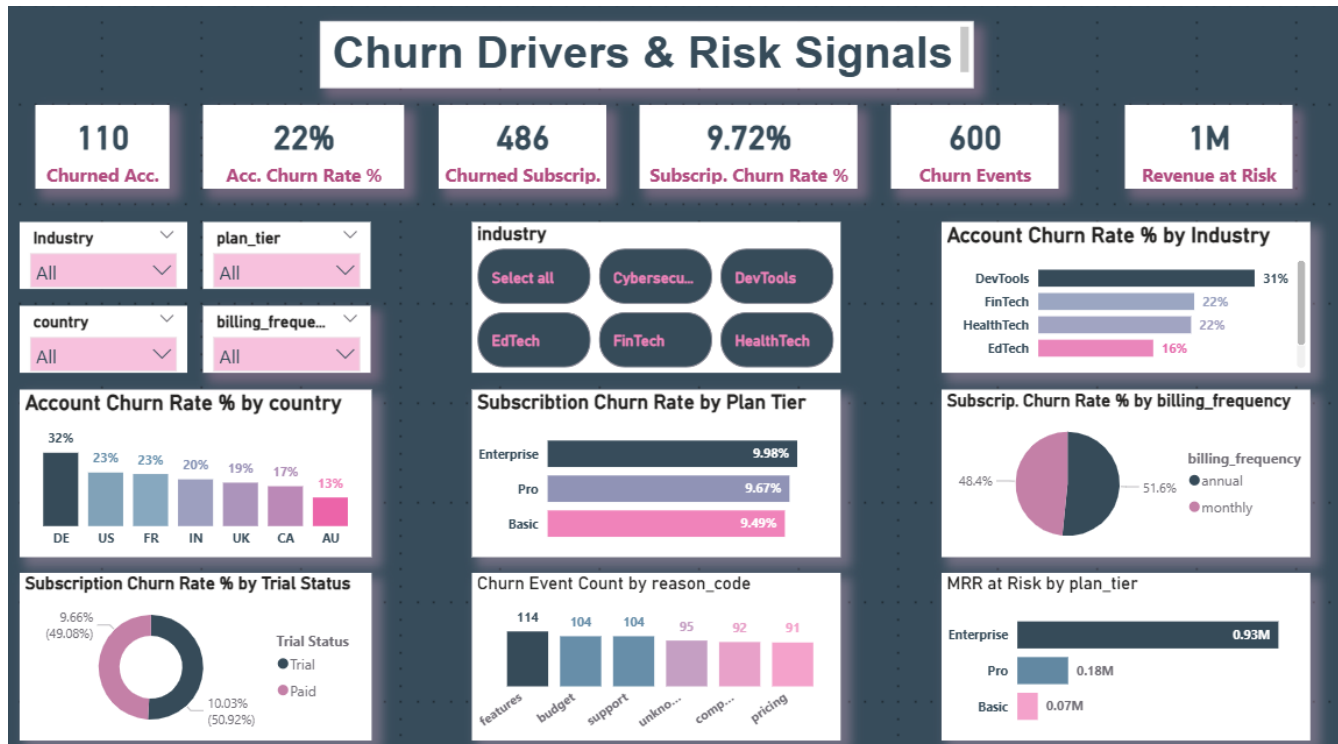
This page provides a high-level summary of customer base size, churn rates, recurring revenue, revenue at risk, and high-level churn exposure across core business segments.





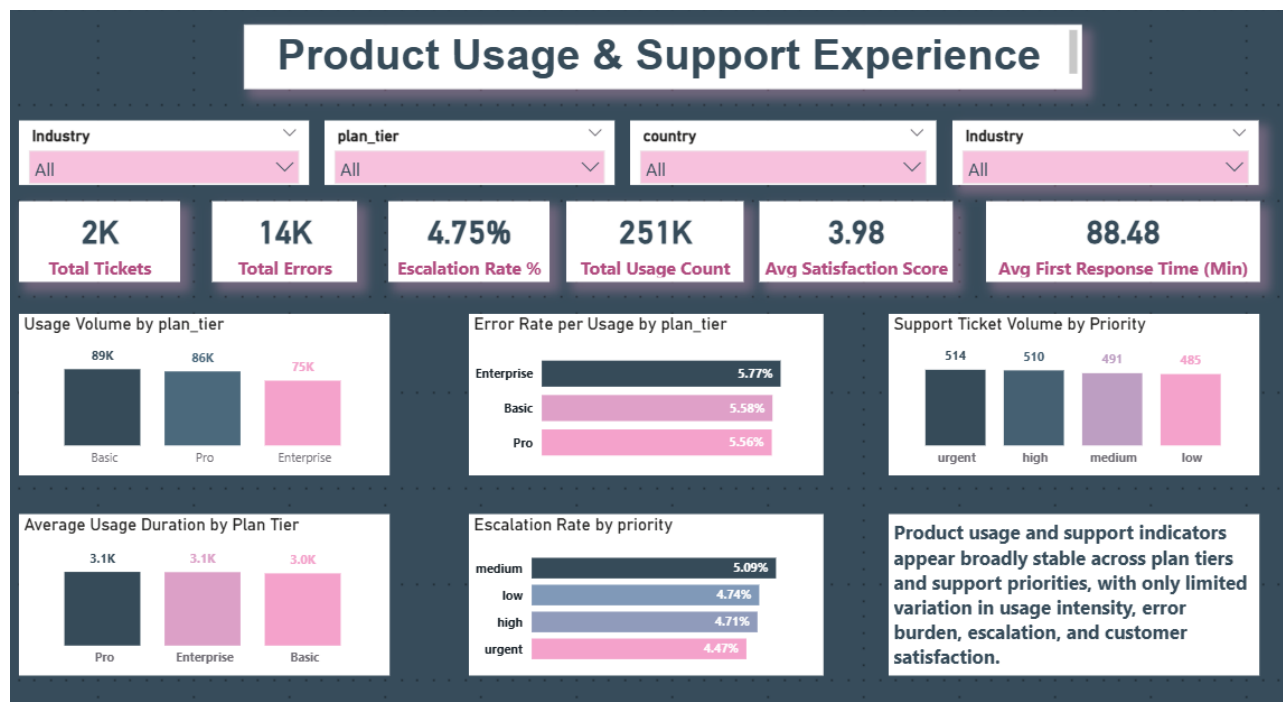
- Churn Drivers and Risk Signals**

This page focuses on churn-related segmentation and interpretation by highlighting patterns across plan tiers, billing structure, customer groups, and recorded churn reasons.



- Product Usage and Support Experience**

This page presents operational context related to product engagement, product friction, support burden, escalation behavior, and customer satisfaction in order to support churn interpretation through behavioral and service-related signals.





The main value of the Power BI layer lies in its ability to convert the analytical outputs of the project into a format that is easier for business teams to consume. Through interactive filtering and visual summaries, the dashboards make it possible to examine churn and retention from multiple perspectives without requiring direct access to the notebook or feature-engineering workflow.

From a stakeholder perspective, this reporting layer supports several important use cases:

- executive visibility into retention health and revenue exposure
- customer success prioritization through churn-oriented segmentation
- product and support review through operational indicators
- interactive exploration of churn-related business patterns
- communication of findings in a more decision-oriented format

Although the predictive modeling results remained weak, the dashboard layer still adds substantial value to the project because the dataset proved to be more useful for **descriptive and diagnostic churn analysis** than for high-confidence predictive classification. In that context, Power BI serves as the most practical final delivery format for transforming the project into a usable **retention intelligence framework**.

Overall, the Power BI reporting layer strengthens the business applicability of the project by linking technical analysis to visual decision support. It helps position the work not only as a data science exercise, but also as a practical analytics solution that can support customer retention monitoring, revenue-risk awareness, and cross-functional business communication.